Lean check of the BINSTD n=17 factor product

Validation audit, 22 October 2026

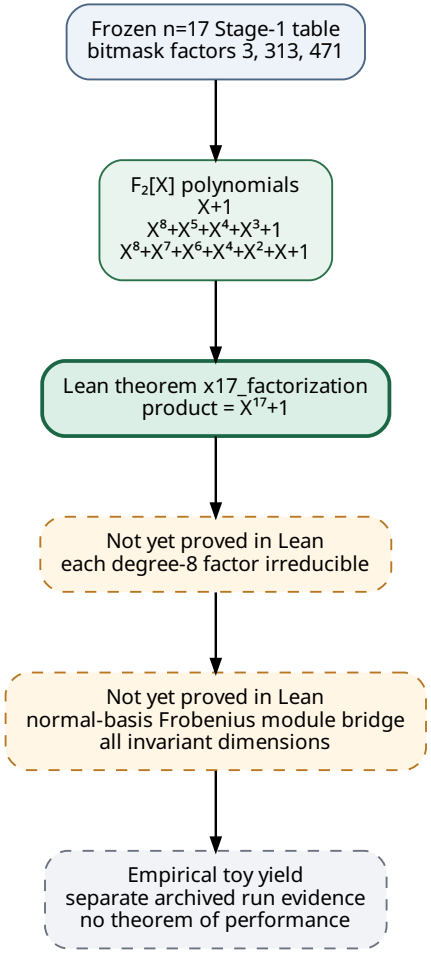
Frozen source. The BINSTD Stage-1 admissible-dimension files agree byte for byte and list bitmask polynomials 3, 313 and 471. The source-binding checker converts the Lean definitions back to these exact bitmasks; the frozen file SHA-256 begins `6f09e6c620fc`.

Lean theorem. In $(\mathbf{Z}/2\mathbf{Z})[X]$,

$$(X+1)(X^8+X^5+X^4+X^3+1)(X^8+X^7+X^6+X^4+X^2+X+1) = X^{17}+1.$$

The proof expands the product and reduces all even coefficients using characteristic two. The formal task names the exact theorem; Lean 4.35.0-rc1 reports `machine_verified`, a passing full build, 617 theorem declarations using only standard axioms and a passing target audit. Independent semantic review remains pending.

Boundary. This checks a polynomial product. It does not prove the degree-eight factors irreducible, identify field Frobenius with the normal-basis polynomial module, classify every invariant subspace or establish any empirical relation-yield ratio. The archived toy replay is separate evidence. No performance dashboard value or security claim changes.


Frozen n=17 Stage-1 table
bitmask factors 3, 313, 471
$F_2[X]$ polynomials
X+1
$X^8+X^5+X^4+X^3+1$
$X^8+X^7+X^6+X^4+X^2+X+1$
Lean theorem x17_factorization
product = $X^{17}+1$
Not yet proved in Lean
each degree-8 factor irreducible
Not yet proved in Lean
normal-basis Frobenius module bridge
all invariant dimensions
Empirical toy yield
separate archived run evidence
no theorem of performance